Nischal V Pattedar

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Summary

Analytical and detail-oriented CSE-AI undergraduate with a 9.08 CGPA and internship experience at DRDO, with proficiency in Python, SQL, and data analysis. Experienced in working with real-world datasets, extracting actionable insights, and developing data-driven solutions. Strong foundation in analytical problem-solving with a keen interest in SQL and data analytics, and an ability to apply technical skills to solve practical business problems.

Education

Amrita Vishwa Vidyapeetham **Aug. 2023 – Present**
B.Tech in Computer Science and Engineering (Artificial Intelligence), CGPA: 9.08 *Bengaluru, Karnataka*

BRS Global School **2023**
Senior Secondary (Class XII), 87.2% *Bengaluru, Karnataka*

BRS Global School **2021**
Secondary (Class X), 96.4% *Bengaluru, Karnataka*

Experience

Gas Turbine Research Establishment (GTRE), DRDO **Dec. 2025 – Jan. 2026**
Research Intern *Bengaluru, Karnataka*

- Proposed a post-quantum secure image steganography framework integrating CRYSTALS-Kyber768 key encapsulation with AES-256 hybrid encryption for confidential communication.
- Implemented key-dependent randomized multi-channel embedding to improve resistance against statistical and CNN-based steganalysis attacks.
- Developed an end-to-end secure communication system in Java supporting encrypted message embedding and extraction.

Projects

Esports Tournament Management System **Sept. 2025**

- Developed an esports tournament management platform for tournament discovery, team registration, player management, and registration tracking.
- Designed a relational database with interconnected entities for players, teams, games, tournaments, matches, statistics, and registrations.
- Built an admin dashboard for managing teams, players, and tournaments, with transactional team registration workflows and database integration.

Reddit Sentiment Schism Engine **May. 2026**

- Built a Big Data pipeline to ingest Reddit JSONL dumps from Arctic Shift using Python and Apache Kafka, routing posts across ecosystem-specific topics for downstream processing.
- Developed a PySpark batch processing workflow to parse semi-structured JSON data, apply VADER sentiment analysis, and aggregate sentiment statistics across enthusiast, critic, and public ecosystems.
- Implemented sentiment divergence analysis to detect ecosystem-level schisms using mean sentiment differences, and persisted processed records and summary metrics as Parquet datasets.
- Containerized Kafka and Zookeeper with Docker and developed an interactive Streamlit + Plotly dashboard for sentiment trends, ecosystem comparisons, schism alerts, and sample-post analysis.

Technical Skills

Languages: Python, SQL, JavaScript
Big Data: Apache Kafka, Apache Spark, PySpark, ETL, Batch Processing
Data Science: Pandas, NumPy, Scikit-learn
Databases: PostgreSQL, MongoDB
Tools: Docker, Git, GitHub, VS Code, Playwright, Postman

Certifications & Achievements

AWS Academy Cloud Foundations **Aug. 2025**
Amazon Web Services

1st Place – Google UX Design Challenge **Jan. 2025**
Google Developer Groups, Amrita School of Engineering